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| 3 | <p>(a) The graph is a <u>straight line passing through the origin</u> which shows that the acceleration is directly proportional to the displacement.</p> <p>The straight line has a <u>negative gradient</u> which shows that direction of acceleration is always opposite to the direction of displacement.</p>                                                                                             |  |
|   | <p>(b) When the eardrum travels a distance of 0.50 cm from max displacement, its displacement will be 1.50 cm from equilibrium position.</p> $x = x_0 \cos(\omega t)$ $1.50 = 2.00 \cos\left(\frac{2\pi}{2.10}t\right)$ $t = 0.242 \text{ s}$                                                                                                                                                                 |  |
|   | <p>(c) Total energy = Maximum kinetic energy = <math>\frac{1}{2}mv_{\max}^2 = \frac{1}{2}m\omega^2x_0^2</math></p> <p>Potential energy = Total energy - Kinetic energy</p> $= \frac{1}{2}m\omega^2x_0^2 - \frac{1}{2}m\omega^2(x_0^2 - x^2)$ $= \frac{1}{2}m\omega^2x^2$ $= \frac{1}{2} \times 0.100 \times \left(\frac{2\pi}{2.10}\right)^2 \times (0.75 \times 10^{-2})^2$ $= 2.5 \times 10^{-5} \text{ J}$ |  |